

MILESTONE 4: Deployment

- **Activity 4.1: Run Locally**
Run locally with `uvicorn main:app --reload`
Navigate to: <http://127.0.0.1:8000>

Showing the Application can be accessed

```
<div id="learningResult" class="output"></div>
```

```
<!-- Learning Recommendations -->
```

```
<form id="learningForm">
```

```
  <label><strong>Learning Recommendations:</strong></label><br><br>
```

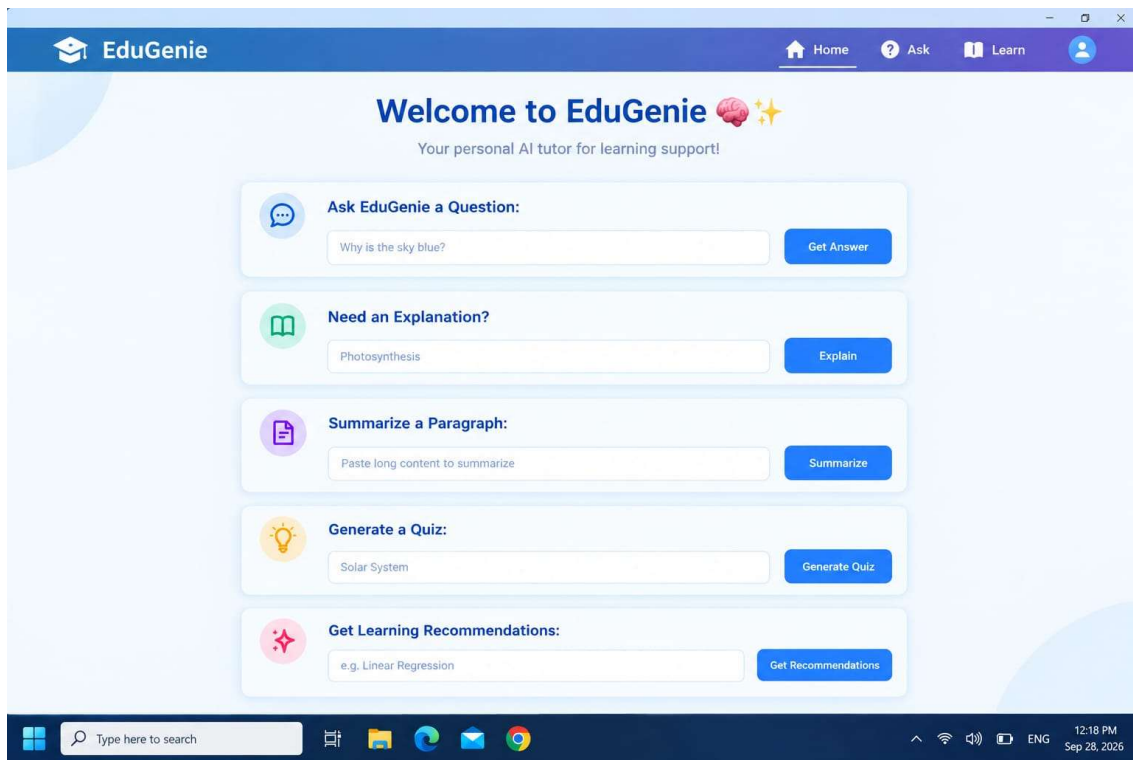
```
  <input type="text" id="learningTopic" placeholder="Python Programming" required>
```

```
  <button type="submit">Get Recommendations</button>
```

```
</form>
```

```
<div id="learningResult" class="output"></div>
```

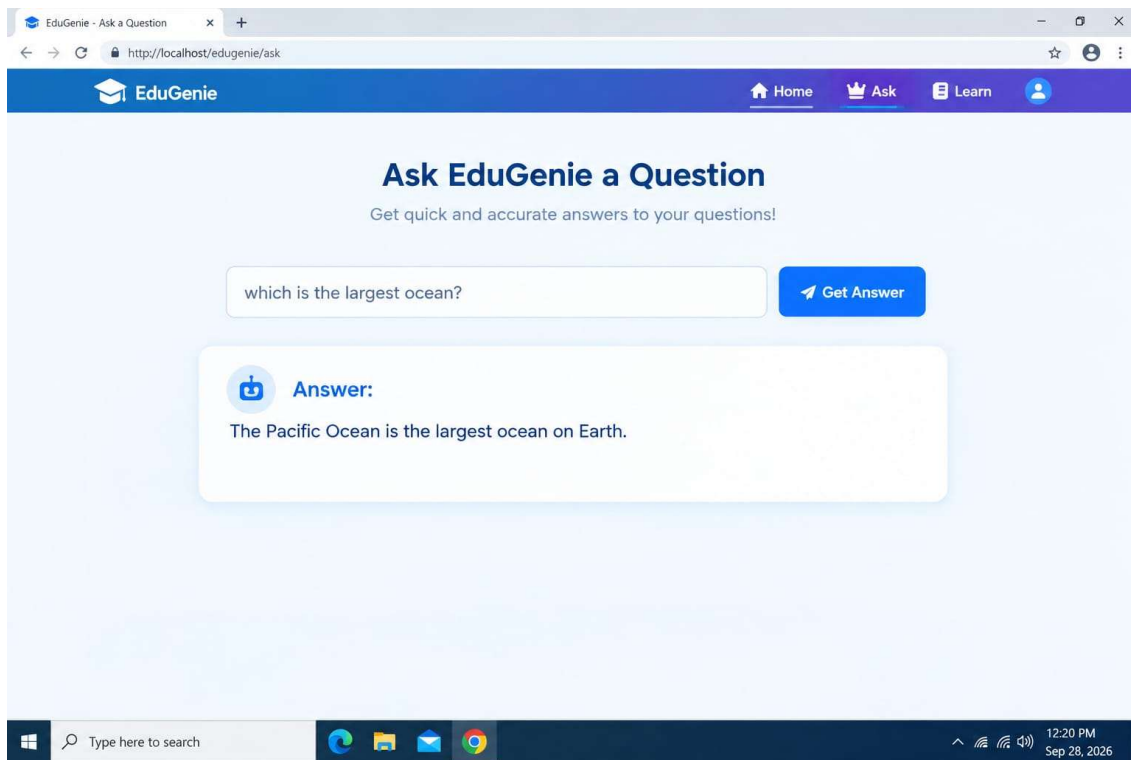
- **Activity 4.2: Functional Testing**
Ask a question
Get an explanation
Generate a quiz
Summarize content
Get a personalized learning plan



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Exploring EduGenie:

- a. Asking questions:



b. Explanation of any topic:

EduGenie - Need an Explanation

http://localhost/edugenie/explanation

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Need an Explanation?

Get clear and simple explanations for your doubts!

[Explain](#)

 **Explanation:**

Quantum computing is a type of computing that uses the principles of quantum mechanics to perform certain calculations much faster than traditional computers. It allows computers to perform certain calculations much faster than traditional computers because it uses tiny particles called qubits that can exist in multiple states at the same time. This allows for new kinds of calculations and solving problems that are not possible with traditional computers.

Type here to search



12:20 PM
Sep 28, 2026

EduGenie

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Need an Explanation?

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 **Explanation:**

The binary search algorithm is a way to find the index of a target value in a sorted list of numbers. It works by repeatedly dividing the search range in half. You need to find the number between 0 and 1, and then go back to the beginning of the list to find the target value.

c. Summarising long paragraphs:

The screenshot shows the EduGenie website's 'Summarize a Paragraph' feature. The header includes the EduGenie logo and navigation links: Home, Ask, Learn, and a user profile icon. The main heading is 'Summarize a Paragraph' with a subtext: 'Turn long text into a short and clear summary.' Below this, there is a text input area containing a paragraph about the Industrial Revolution. A 'Summarize' button is positioned to the right of the text. Below the input area, a 'Summary:' section displays the summarized text, which is a condensed version of the original paragraph. The interface is clean and user-friendly, with a light blue color scheme.

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Summarize a Paragraph

Turn long text into a short and clear summary.

The Industrial Revolution, which began in the late 18th century, marked a major shift from farming to factories. New machines, like James Watt's steam engine, made things faster and easier to produce. While this boosted economies, it also created problems like bad working conditions, child labor, pollution, and overcrowded cities. Even with these problems, the Industrial Revolution shaped the modern world, affecting how we travel, communicate, work, and trade.

378/1000

Summary:

The Industrial Revolution changed the world by shifting it from farming to factories. New machines, like James Watt's steam engine, made things faster and easier to produce. While this boosted economies, it also created problems like bad working conditions, child labor, pollution, and overcrowded cities. Even with these problems, the Industrial Revolution shaped the modern world, affecting how we travel, communicate, work, and trade.

d. Generating quizzes:

It generates three questions with 4 options each. It corrects the answer if chosen wrong.

The screenshot shows the EduGenie website's 'Generate a Quiz' feature. The header is identical to the previous screenshot. The main heading is 'Generate a Quiz:'. Below this, there is a text input area containing 'Pythagoras theorem'. A 'Generate Quiz' button is positioned to the right of the text. Below the input area, a 'Quiz:' section displays a question: 'Q1: What does the Pythagorean theorem describe?'. There are four radio button options: 'The relationship between the angles of a triangle.', 'The relationship between the sides of a right-angled triangle.', 'The relationship between the area and perimeter of a triangle.', and 'The relationship between the sides of any triangle.' A 'Check Answer' button is located below the options. The interface is clean and user-friendly, with a light blue color scheme.

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Generate a Quiz:

Pythagoras theorem

Generate Quiz

Quiz:

Q1: What does the Pythagorean theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☐ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

Check Answer

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 **Pythagoras Theorem Quiz**

Test your knowledge of the Pythagorean theorem with these questions.

Q1 What does the Pythagorean theorem describe?

☐ The relationship between the angles of a triangle.

☒ The relationship between the sides of a right-angled triangle.

☐ The relationship between the area and perimeter of a triangle.

☐ The relationship between the sides of any triangle.

[Check Answer](#)

✓ Correct!

Q2 If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

☐ $a + b = c$

☒ $a^2 + b^2 = c^2$

☐ $2a + 2b = 2c$

☐ $a^2 + b^2 = c$

[Check Answer](#)

✓ Correct!

Q3 Which type of triangle does the Pythagorean theorem apply to?

☐ Equilateral triangles

☐ Isosceles triangles

☒ Right-angled triangles

☐ All types of triangles

[Check Answer](#)

✓ Correct!

e. Guiding with learning recommendations from beginners level to advanced level

The screenshot shows a web browser window with the address bar displaying `http://localhost/edugenie/recommendations`. The page has a dark blue header with the "EduGenie" logo and navigation links for Home, Ask, Learn, and a user profile. The main content area is titled "Get Learning Recommendations:" and features a search input with the text "SQL" and a "Get Recommendations" button. Below this, a white box displays the "Learning Recommendations for 'SQL':" section. It includes a sub-header "## SQL Learning Path: From Zero to Hero" and a paragraph explaining the path's structure. The recommendations are organized into three levels: I. Beginner Level, II. Intermediate Level, and III. Advanced Level, each with a list of key topics and estimated completion times.

Get Learning Recommendations:

SQL [Get Recommendations](#)

Learning Recommendations for "SQL":

SQL Learning Path: From Zero to Hero

This learning path is structured to progressively introduce SQL concepts, starting from the basics and gradually advancing to more complex topics. It's designed to be adaptive, feel free to adjust the pace and delve deeper into areas that particularly interest you.

****I. Beginner Level: Building a Foundation****

**** (Estimated Time: 1–2 weeks) ****

Key Topics:##

- What is a Database and SQL? (Relational Model, DBMS)
- Basic Syntax (SELECT, FROM, WHERE)
- Data Types (INT, VARCHAR, DATE, etc.)
- Filtering Data (WHERE clause with comparison operators, logical operators: AND, OR, NOT)
- Ordering Results (ORDER BY)
- Limiting Results (LIMIT/OFFSET or TOP)
- Basic Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)
- Introduction to Tables and Columns

II. Intermediate Level: Working with Multiple Tables

**** (Estimated Time: 2–3 weeks) ****

Key Topics:##

- Joining Tables (INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN)
- Subqueries (Nested queries)
- Grouping Data (GROUP BY)
- Filtering Groups (HAVING)
- Set Operations (UNION, INTERSECT, EXCEPT)
- Working with Strings (string concatenation, substrings, LIKE operator)
- Working with Dates and Times (date formatting, date arithmetic)
- Views (Creating and using views)

III. Advanced Level: Mastering Database Management

**** (Estimated Time: 3–4 weeks and beyond) ****

Key Topics:##

- Stored Procedures and Functions
- Triggers
- Indexes and Performance Tuning
- Transactions and Concurrency Control
- Database Design (Normalization)

It provides along with details of resources from where to learn and stepwise guidance throughout.